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**Section C**

You are advised to spend no more than **60 minutes** on this section.

Enter your answers to **Section C** in your Electronic Answer Document. You **must save** this document at regular intervals.

These questions require you to load the **Skeleton Program** and to make programming changes to it.

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**1 4**

This question refers to the subroutine `SelectMove`. This subroutine makes three calls to the subroutine `DisplayErrorCode` to notify the user of errors. The error codes passed as parameters are to be made more informative for the user.

**What you need to do:**

**Task 1**

Modify the subroutine `DisplayErrorCode` so that meaningful error messages are displayed for each type of error explaining the circumstances which caused each error. The error code should also be displayed.

**Task 2**

Test that the changes you have made work by conducting the following test:

- run the program
- enter Y
- load `game1.txt`
- enter the string `a4`
- enter the string `a9`
- enter 3
- enter 4
- enter a
- enter 9
- enter 3
- enter 0

**Evidence that you need to provide**

Include the following evidence in your Electronic Answer Document.

**1 4 . 1**

Your PROGRAM SOURCE CODE for the entire subroutine `DisplayErrorCode`.  
**[3 marks]**

**1 4 . 2**

SCREEN CAPTURE(S) showing the requested test including the list of possible moves for Player A.

**[1 mark]**

**1 5**

This question refers to the subroutine `ValidJump`. The rules of the game are to be amended. Instead of jumping over their own piece, a player can only jump over an opponent's piece.

**What you need to do:**

**Task 1**

Amend the subroutine `ValidJump` so that a jump is only possible if the middle piece belongs to the opponent.

**Task 2**

Test that the changes you have made work by conducting the following test:

- run the program
- enter Y
- load `game3.txt`
- enter the string `a5`
- enter 5
- enter 0

**Evidence that you need to provide**

Include the following evidence in your Electronic Answer Document.

**1 5****1**

Your PROGRAM SOURCE CODE for the entire subroutine `ValidJump`.

**[2 marks]**

**1 5****2**

SCREEN CAPTURE(S) showing the requested test including the list of possible moves **before** the jump and your test input, and then the board display **after** the jump.

**[1 mark]**

**Turn over for the next question**

**Turn over ►**

**1 6**

This question refers to the subroutine `PrintResult`. The rules of the game are to be amended. The game still ends when a player cannot make a move, but the winner is to be determined using a formula that calculates a score for each player. The winner is the player with the **lowest** score. The formula to calculate the score is shown in **Figure 5**.

**Figure 5**

Player's score = Number of that player's moves **minus** total number of player's pieces on the board **minus** number of that player's dames **multiplied by** 10

For example, if Player A made 40 moves and has 12 pieces on the board, and 3 of them are dames, then Player A's score is  $40 - 12 - (3 \times 10) = -2$

Note that a dame is also considered to be a piece.

**What you need to do:****Task 1**

Create a new subroutine `CountNumberOfPieces`. This subroutine is to count the number of pieces a player has on the board.

**Task 2**

Amend the subroutine `PrintResult` to implement the formula given in **Figure 5** and output the score for each player and display the winner.

You must consider the possibility of a draw.

**Task 3**

- run the program
- enter Y
- load `game4.txt`

**Evidence that you need to provide**

Include the following evidence in your Electronic Answer Document.

**1 6****1**

Your PROGRAM SOURCE CODE for the entire subroutine `CountNumberOfPieces` and the entire subroutine `PrintResult`.

**[9 marks]****1 6****2**

SCREEN CAPTURE(S) showing all the output from the requested test including the board display.

**[1 mark]**

|   |   |
|---|---|
| 1 | 7 |
|---|---|

This question will further change the rules of the game.

When a piece is promoted to a dame, the player who the new dame belongs to now chooses **one** of the opponent's pieces. This piece is removed from the board and the dame is placed in the square the removed piece was in.

|   |   |   |
|---|---|---|
| 1 | 7 | 1 |
|---|---|---|

State the identifier of the data structure that now needs to be passed as a parameter into the subroutine `MoveDame`.

[1 mark]

**What you need to do:**

**Task 1**

Amend the subroutine `MoveDame`.

This subroutine is to:

- ask the player the piece ID of the opponent's piece they want to remove
- check that the piece is an opponent's piece and is on the board
- remove the opponent's piece
- return the coordinates for the new dame.

**Task 2**

Amend the calls to `MoveDame` from within the subroutine `MovePiece`.

You will need to amend the parameter list of the subroutine heading of `MovePiece` and the call to `MovePiece` from within the subroutine `MakeMove`.

**Task 3**

Test that the changes you have made work by conducting the following test:

- run the program
- enter Y
- load `game3.txt`
- move a2 to row 7, column 0
- take piece b1

**Task 4**

- move b5 to row 0, column 3
- take piece a6

**Question 17 continues on the next page**

**Turn over ►**

**Evidence that you need to provide**

Include the following evidence in your Electronic Answer Document.

- 1 7 . 2** Your PROGRAM SOURCE CODE for the entire subroutine `MoveDame` and the entire subroutine `MakeMove`. **[9 marks]**
- 1 7 . 3** SCREEN CAPTURE(S) showing the requested test including the board display after piece `b1` has been taken. **[1 mark]**
- 1 7 . 4** SCREEN CAPTURE(S) showing the requested test including the board display after piece `a6` has been taken. **[1 mark]**

**END OF QUESTIONS**